

Outline { partial trace, reduced density operator  
quantum operation (§3.5) '20.11.06.  
(~ §3.3)

Thm 3.15 & Def 3.16 (partial trace)

Let  $\mathcal{H}^A, \mathcal{H}^B$ : Hilbert spaces with ONBs  $\{|e_a\rangle\}$  and  $\{|f_b\rangle\}$ , and let  $M \in L(\mathcal{H}^A \otimes \mathcal{H}^B)$  and  $M_{a_1 b_1, a_2 b_2}$  be the matrix of  $M$  in ONB  $\{|e_{a_1} \otimes f_{b_1}\rangle\}$ .

Define  $\begin{cases} \underline{\text{tr}^B(M)} := \sum_{a_1, a_2, b} M_{a_1 b, a_2 b} |e_{a_1}\rangle \langle e_{a_2}| \in L(\mathcal{H}^A) \\ \underline{\text{tr}^A(M)} := \sum_{b_1, b_2, a} M_{a b_1, a b_2} |f_{b_1}\rangle \langle f_{b_2}| \in L(\mathcal{H}^B). \end{cases}$

Then, we have

(i)  $\text{tr}^B(M)$  and  $\text{tr}^A(M)$  do not depend on the choice of ONBs.

(ii)  $\text{tr}^B(M)$  and  $\text{tr}^A(M)$  are unique in the sense that

$$\begin{cases} \text{tr}(M^A \text{tr}^B(M)) = \text{tr}((M^A \otimes \mathbb{1}^B) M) \quad \forall M^A \in L(\mathcal{H}^A) \\ \text{tr}(M^B \text{tr}^A(M)) = \text{tr}((\mathbb{1}^A \otimes M^B) M) \quad \forall M^B \in L(\mathcal{H}^B). \end{cases}$$

(3.47)

Here,  $\text{tr}^B$  is called as the partial trace over  $\mathcal{H}^B$  and  $\text{tr}^A$  is called as the partial trace over  $\mathcal{H}^A$ .

(sketch of the proof)

(i) (independence on ONB). Let  $\{|e_a\rangle\}$  and  $\{|\tilde{f}_b\rangle\}$  be two other ONBs. Then,  $\exists U^A \in \mathcal{U}(\mathcal{H}^A), U^B \in \mathcal{U}(\mathcal{H}^B)$  s.t.  $|\tilde{e}_a\rangle = U^A |e_a\rangle$  and  $|\tilde{f}_b\rangle = U^B |f_b\rangle$ .

(ii) Verification of (3.47):  $M \in L(\mathcal{H}^A \otimes \mathcal{H}^B)$

$$M = \sum_{a_1, a_2, b_1, b_2} M_{a_1 b_1, a_2 b_2} |e_{a_1} \otimes f_{b_1}\rangle \langle e_{a_2} \otimes f_{b_2}|.$$

For  $M^A \in L(\mathcal{H}^A)$ ,  $M^A = \sum_{a_1, a_2} M^A_{a_1, a_2} |e_{a_1}\rangle \langle e_{a_2}|$

(iii) (Uniqueness) Let  $\widetilde{\text{tr}^B(M)}$  be another operator on  $\mathcal{H}^A$  satisfying (3.47)<sub>1</sub>.

Then,  $\text{tr} \left( \underbrace{M^A}_{\text{red circle}} \underbrace{(\widetilde{\text{tr}^B(M)} - \text{tr}^B(M))}_{\text{red wavy line}} \right) = 0$  □

Remark (i)  $\underbrace{\text{tr}(M^A \text{tr}^B(M))}_{\in L(\mathcal{H}^A)} = \text{tr} \left( \underbrace{(M^A \otimes \mathbb{1}^B)}_{\in L(\mathcal{H}^A \otimes \mathcal{H}^B)} M \right)$

Thus,  $\text{tr}^B(M)$  is a unique operator satisfying the relation above.

(ii)  $\text{tr} : L(\mathcal{H}) \rightarrow \mathbb{C}$  : complex number. ; "trace"

$\text{tr}^B : L(\mathcal{H}^A \otimes \mathcal{H}^B) \rightarrow L(\mathcal{H}^A)$  ; "partial trace"

Exercise 3.42 For any  $M \in \mathcal{L}(\mathcal{H}^A \otimes \mathcal{H}^B)$ ,  
 $\text{tr}(\text{tr}^B(M)) = \text{tr}(M) = \text{tr}(\text{tr}^A(M))$ .

Thm 3.17 & Def 3.18 (Reduced density operator).

Let  $\rho \in \mathcal{D}(\mathcal{H}^A \otimes \mathcal{H}^B)$  be a density operator on  $\mathcal{H}^A \otimes \mathcal{H}^B$ .

(i) Then,  $\rho^A(\rho) := \text{tr}^B(\rho)$  is the uniquely determined density operator.

(ii) For any bounded observable  $M^A \in \mathcal{B}_{sa}(\mathcal{H}^A)$ ,  
 (set of bounded and self-adjoint operator on  $\mathcal{H}^A$ )  
 $\langle M^A \rangle_{\rho^A(\rho)} = \langle M^A \otimes \mathbb{1}^B \rangle_{\rho}$ .

(iii)  $\{|e_a\rangle\}, \{|f_b\rangle\}$  : ONBs in  $\mathcal{H}^A$  and  $\mathcal{H}^B$ .

$P_{a_1 b_1, a_2 b_2}$  : matrix of  $\rho$  in ONB  $\{|e_a \otimes f_b\rangle\}$  in  $\mathcal{H}^A \otimes \mathcal{H}^B$ .

Then,  $\rho^A(\rho)_{a_1 a_2} = \sum_b P_{a_1 b, a_2 b}$  — (3.52)

Here,  $\rho^A(\rho)$  is called as the reduced density operator.

(Sketch of the proof)

(i) ( $\rho^A(\rho)$  : density operator)

- self-adj. :  $(\rho^A(\rho)_{a_1 a_2})^* = \rho^A(\rho)_{a_1 a_2}$

$\because (\rho^A(\rho)_{a_1 a_2})^* = \frac{\rho^A(\rho)_{a_2 a_1}}{\rho^A(\rho)_{a_2 a_1}} = \sum_b P_{a_2 b, a_1 b}$

$(\rho = \rho^*)$   
 $= \sum_b P_{a_1 b, a_2 b} = \rho^A(\rho)_{a_1 a_2}$

- positivity: For any  $|\psi\rangle \in \mathbb{H}^A$ ,

$$\langle \psi | \rho^A(p) | \psi \rangle = \sum_{a_1, a_2} \overline{\psi_{a_1}} \rho^A(p)_{a_1 a_2} \psi_{a_2}$$

$$= \sum_{a_1, a_2, b} \overline{\psi_{a_1}} \rho_{a_1 b, a_2 b} \psi_{a_2} = \sum_b \langle \psi \otimes f_b | \rho(\psi \otimes f_b) \rangle \geq 0.$$

$$\begin{aligned} - \operatorname{tr}(\rho^A) &= 1; \quad \operatorname{tr}(\rho^A(p)) = \operatorname{tr}(\operatorname{tr}^B(p)) \\ &= \operatorname{tr}(p) = 1. \end{aligned}$$

$$(ii) \langle M^A \otimes \mathbb{1}^B \rangle_p = \operatorname{tr}((M^A \otimes \mathbb{1}^B)p) = \operatorname{tr}(M^A \operatorname{tr}^B(p))$$

$$= \operatorname{tr}(M^A \rho^A(p)) = \langle M^A \rangle_{\rho^A(p)}$$

( $\therefore$  The quantum mechanical expectation value of an observable  $A$  in a mixed state  $p$   $\langle A \rangle_p := \operatorname{tr}(pA)$ .)

Remark  $\rho^A(p)$  is the state obtained from the state  $p \in \mathcal{D}(\mathbb{H}^A \otimes \mathbb{H}^B)$  by taking partial trace over subsystem  $B$ . ;  $\rho^A(p) = \operatorname{tr}^B(p)$

Exercise 3.43.  $M^A \in L(\mathbb{H}^A)$ ,  $M^B \in L(\mathbb{H}^B)$ .

$$M^A \otimes M^B \in L(\mathbb{H}^A \otimes \mathbb{H}^B)$$

$$(i) \operatorname{tr}(M^A \otimes M^B) = \operatorname{tr}(M^A) \cdot \operatorname{tr}(M^B)$$

$$(ii) \operatorname{tr}^B(M^A \otimes M^B) = \operatorname{tr}(M^B) M^A$$

$$(iii) \operatorname{tr}^A(M^A \otimes M^B) = \operatorname{tr}(M^A) M^B.$$

Exercise 3.45  $V$ : finite dim'd v-sp. over a field  $F$ .

$$L(V^{\otimes n}) = L(V)^{\otimes n}$$



$\mathbb{H}$ : qubit Hilbert space  $\simeq \mathbb{C}^2$ .

a system of  $n$  qubits is described  $\mathbb{H}^{\otimes n}$ .

In general, any observable of such system or its operator on  $n$  qubits.

$\iff$  any such operator can be represented as a linear combination of  $n$ -fold tensor products of operators on  $\mathbb{H}$ .

Remark This does not mean that every  $A \in L(V^n)$  is of the form

$$A = A_1 \otimes \cdots \otimes A_n \quad \text{for some } \{A_i\} \in L(V).$$

$$A = \sum_j a_{j_1, \dots, j_n} A_{j_1} \otimes \cdots \otimes A_{j_n} \quad \{A_{j_i}\} \in L(V).$$

(sketch of the proof).

claim (i)  $L(V^{\otimes n}) \subseteq L(V)^{\otimes n}$       (ii)  $\dim(L(V^{\otimes n})) = \dim(L(V)^{\otimes n})$

(i) Any  $w \in V^{\otimes n}$ ,  $w = \sum w_{j_1, \dots, j_n} v_{j_1} \otimes \cdots \otimes v_{j_n}$

$\{v_j\}_{j=1}^{\dim V}$  : basis of  $V$ .

For any  $A_1, \dots, A_n \in L(V)$ ,

$$(A_1 \otimes \cdots \otimes A_n)w = \sum_{j_1, \dots, j_n} w_{j_1, \dots, j_n} (A_1 v_{j_1}) \otimes \cdots \otimes (A_n v_{j_n}).$$

$$\Rightarrow A_1 \otimes \dots \otimes A_n \in L(V^{\otimes n})$$

$$\text{ii)} \quad \dim L(V) = (\dim V)^2, \quad \dim (V^{\otimes n}) = (\dim V)^n$$

$$\Rightarrow \dim (L(V)^{\otimes n}) = (\dim L(V))^n = (\dim V)^{2n}$$

$$\dim L(V^{\otimes n}) = (\dim V^{\otimes n})^2 = (\dim V)^{2n}.$$

### § 3.5 Quantum operation

There are two ways that a quantum system can change

(i) by unitary time evolution by some Hamiltonian

(ii) a state transformation by a measurement

#### Motivation

1. For a system A of our interest,  $\rho^A \in \mathcal{D}(\mathcal{H}^A)$ .

2. Combine  $(A, \rho^A)$  with the environment  $(B, \rho^B)$ .

$$\iota_{\rho^B} : \mathcal{D}(\mathcal{H}^A) \rightarrow \mathcal{D}(\mathcal{H}^A \otimes \mathcal{H}^B)$$

$$\rho^A \mapsto \rho^A \otimes \rho^B$$

3.  $U \in \mathcal{U}(\mathcal{H}^A \otimes \mathcal{H}^B)$ ;  $U : \mathcal{D}(\mathcal{H}^A \otimes \mathcal{H}^B) \rightarrow \mathcal{D}(\mathcal{H}^A \otimes \mathcal{H}^B)$

$$\rho^A \otimes \rho^B \mapsto \underline{U(\rho^A \otimes \rho^B)U^\dagger}$$

4.  $P^B$ : projection onto the eigenspace of an observable of B.

$$\square = U(\rho^A \otimes \rho^B)U^\dagger \mapsto \frac{(\mathbb{1}^A \otimes P^B) \square (\mathbb{1}^A \otimes P^B)}{\text{tr}((\mathbb{1}^A \otimes P^B) \square)}$$

Note This measurement

$$\rho \xrightarrow{\text{state before measurement}} \frac{P \rho P}{\text{tr}(P \rho)} : \text{state after measurement.}$$

5. Finally, ignoring  $B$ , we can obtain a description of syst.  $A$  by taking the partial trace over  $B$ .

$$\frac{(\mathbb{1}^A \otimes P^B) \rho (\mathbb{1}^A \otimes P^B)}{\text{tr}((\mathbb{1}^A \otimes P^B) \rho)} \xrightarrow{\text{partial trace over } B} \frac{\text{tr}^B((\mathbb{1}^A \otimes P^B) \rho (\mathbb{1}^A \otimes P^B))}{\text{tr}((\mathbb{1}^A \otimes P^B) \rho)}$$

$\in D(\mathcal{H}^A \otimes \mathcal{H}^B)$   
 $\in D(\mathcal{H}^A)$

$\rho^A$

Our goal is to represent  $(*)$  in terms of suitable operators on  $\mathcal{H}^A$  only.

Lemma 3.23

$\mathcal{H}$ : Hilbert,  $\{K_\ell\}_{\ell=1}^m \subseteq \mathcal{L}(\mathcal{H})$ . Then, for any

$\rho \in D(\mathcal{H})$ ,

the operator  $K(\rho) = \sum_{\ell=1}^m K_\ell \rho K_\ell^*$  satisfies

(i)  $K(\rho)^* = K(\rho)$ : self-adj.

(ii)  $K(\rho) \geq 0$ : positive.

(iii) Moreover, for  $\delta \in (0, 1]$  and  $\rho \in D(\mathcal{H})$ .

$$\sum_{\ell=1}^m K_\ell^* K_\ell \leq \delta \mathbb{1} \iff \text{tr}(K(\rho)) \leq \delta$$

Thm 3.24 & Def 3.26 (Quantum operation).

$H^A$ : finite dim'l Hilbert,  $K: D(H^A) \rightarrow \underbrace{D(H^A)}_1$

$$\{ \rho \in D(H^A) : \rho^* = \rho, \rho \geq 0, \text{tr}(\rho) \leq 1 \}.$$

Then, for  $\delta \in (0, 1]$ , TPAE

(i)  $\exists H^B$ : finite dim'l,  $V \in B(H^A \otimes H^B)$ ,  $\rho^B \in D(H^B)$   
s.t.  $V^* V \leq \delta \mathbb{1}^{AB}$  and  $K(\rho^A) = \text{tr}^B (V(\rho^A \otimes \rho^B) V^*)$   
(3.74)

(ii)  $\exists \{K_\ell\}_{\ell=1}^m \subseteq L(H^A)$  s.t.

$$\sum_{\ell=1}^m K_\ell^* K_\ell \leq \delta \mathbb{1}^A \text{ and } K(\rho^A) = \sum_{\ell=1}^m K_\ell \rho^A K_\ell^* \quad (3.76)$$

Here, a quantum operation  $K$  is a convex-linear map that can be represented by (3.74) or (3.76).

Moreover, <sup>(i)</sup> if  $K(\rho) = \sum_{\ell=1}^m K_\ell \rho K_\ell^*$  where  $\{K_\ell\} \subseteq L(H)$

with  $\sum_{\ell=1}^m K_\ell^* K_\ell \leq \mathbb{1}$ , then  $\{K_\ell\}$ : Kraus operators.

is called as the operator-sum representation.

NOT unique

(ii)  $K(\rho) = \text{tr}^B (V(\rho \otimes \rho^B) V^*)$  with  $V^* V \leq \mathbb{1}$

is called as the environment representation

Remark (i) Thm 3.24 does not assert that every convex-linear map does not have one of the equivalent forms (3.174) or (3.176).

- (ii) environments rep.  $\longleftrightarrow$  operator-sum rep.
- concrete and easy to related real-world
  - Mathematically inconvenient
  - rather abstract
  - useful for calculations.

Cor 3.27

$K: D(H) \rightarrow D_{\leq}(H)$  : quantum operation with Kraus operators  $\{K_\ell\}_{\ell=1}^m$ .

For  $\tilde{m} \geq m$  and  $U \in U(\tilde{m})$  : unitary matrix.

$\tilde{K}_j := \sum_{\ell=1}^m U_{j\ell} K_\ell$  ;  $\{\tilde{K}_j\}_{j=1}^{\tilde{m}}$  : Kraus operators

Summary

1. initially  $\rho \in D(H)$
2. combine with another system (environment, error)
3. subject to  $\left\{ \begin{array}{l} \text{unitary time evolution} \\ \text{possible measurement.} \end{array} \right.$

4.  $D(H) \rightarrow D(H)$   
 $\rho \mapsto \frac{K(\rho)}{\text{tr}(K(\rho))}$  : with the help of a quantum operation  $K$ .

Reference

[Quantum computation and quantum information, Nielsen and Chuang] ch 8. Quantum-noise and quantum operations.